

ABURATSU, Japan

WMO: 478350

Lat: 31.5786N

Lon: 131.4078E

Elev: 48

StdP: 14.67

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	34.4	36.6	17.7	13.4	41.3	20.5	15.4	42.2	28.7	49.1	25.6	50.8	10.0	290	0.439

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.6	91.1	78.7	89.3	78.9	87.7	78.6	80.9	87.0	80.1	86.1	79.4	85.2	11.8	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.3	152.3	84.3	78.5	148.1	83.7	77.7	144.5	83.2	44.6	87.2	43.8	86.5	43.0	85.4	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
26.0	21.4	18.6	DB	30.3	94.9	1.6	2.0	29.1	96.3	28.2	97.5	27.3	98.6	26.1	100.0	
			WB	26.5	82.0	1.9	1.0	25.1	82.7	24.0	83.3	22.9	83.8	21.6	84.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	65.5	48.2	50.2	55.1	62.2	69.1	74.2	81.5	82.7	78.2	70.2	61.0	51.9
	DBStd	12.70	5.03	5.90	5.43	4.69	3.44	4.21	3.20	2.36	3.75	4.55	5.73	5.66
	HDD50	217	95	63	14	0	0	0	0	0	0	0	1	44
	HDD65	1918	521	415	309	106	6	0	0	0	0	9	146	406
	CDD50	5864	39	69	173	368	593	727	976	1012	845	628	330	104
	CDD65	2087	0	1	2	23	134	277	511	547	395	172	25	0
	CDH74	17519	0	0	2	47	401	1546	5362	6130	3302	692	37	0
	CDH80	5224	0	0	0	3	32	297	1778	2231	820	61	2	0
Wind	WSAvg	9.3	10.2	9.9	9.9	9.5	8.7	8.4	8.6	8.8	9.0	9.3	9.3	9.8
Precipitation	PrecAvg	102.8	3.3	4.7	7.4	8.9	9.2	19.6	11.3	10.7	12.6	7.2	4.6	3.2
	PrecMax	166.1	10.0	11.9	14.5	17.8	18.3	34.1	41.3	25.7	29.4	19.4	14.3	8.5
	PrecMin	79.6	1.1	1.5	3.4	2.1	4.8	9.4	1.6	1.1	2.7	0.7	1.3	0.1
	PrecStd	18.7	2.0	2.9	2.8	3.9	3.3	6.5	8.9	6.1	7.6	4.5	3.3	2.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.8	70.6	72.9	78.9	83.6	89.3	93.7	94.3	90.6	84.3	77.8	70.2
		MCWB	59.1	62.9	60.9	64.3	67.5	77.3	78.6	79.1	77.8	74.4	70.8	62.2
	2%	DB	63.1	66.6	69.6	74.7	80.2	86.1	91.1	91.4	87.8	81.7	74.6	66.9
		MCWB	56.7	60.8	60.4	64.5	68.2	76.6	78.6	79.0	78.0	73.3	66.6	60.0
	5%	DB	60.4	63.8	67.3	72.5	77.9	83.2	89.2	89.6	85.9	79.6	72.6	64.4
		MCWB	53.4	58.0	59.3	63.7	67.7	75.6	78.8	79.3	77.0	71.5	65.0	57.4
	10%	DB	57.7	60.7	65.2	70.7	76.1	81.0	87.4	88.0	84.3	77.8	70.4	61.8
		MCWB	50.7	54.2	58.0	62.5	67.4	74.8	78.5	79.1	76.1	69.9	63.3	54.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	62.1	64.6	66.1	70.7	74.7	79.0	81.9	82.0	81.0	78.0	71.9	65.4
		MCDB	65.4	68.5	69.2	74.4	78.7	85.0	88.2	88.0	86.0	82.1	76.1	67.9
	2%	WB	58.6	62.0	63.4	68.5	72.8	77.9	80.7	81.1	79.7	75.8	69.1	61.7
		MCDB	62.0	66.1	67.2	71.7	76.3	83.4	87.0	87.2	84.7	79.8	73.1	65.4
	5%	WB	55.0	58.9	61.4	66.4	71.2	76.9	79.8	80.4	78.7	73.9	66.9	58.7
		MCDB	59.0	63.1	65.9	70.4	74.7	81.9	86.0	86.6	83.6	77.9	70.9	63.1
	10%	WB	51.4	55.4	59.1	64.6	69.7	75.9	79.0	79.8	77.8	71.8	64.7	56.0
		MCDB	56.5	59.5	63.8	69.1	73.7	80.5	85.1	85.8	82.7	76.2	68.9	60.6
Mean Daily Temperature Range	5% DB	MDBR	15.0	14.7	14.7	14.3	12.9	9.7	10.3	10.6	10.8	12.2	13.9	15.0
		MCDBR	17.4	16.3	16.9	16.7	15.9	12.3	12.9	12.9	12.0	12.9	14.1	16.2
		MCWBR	12.5	12.4	11.5	10.5	9.2	5.7	5.3	5.5	5.9	7.4	9.1	11.2
	5% WB	MCDBR	14.7	14.5	13.8	12.1	10.7	10.0	11.2	11.1	10.3	9.9	10.9	13.7
		MCWBR	12.9	12.5	11.6	9.4	8.1	5.3	5.4	5.5	5.6	6.6	8.6	11.2
Clear-Sky Solar Irradiance	taub		0.426	0.472	0.534	0.555	0.564	0.565	0.543	0.531	0.517	0.491	0.441	0.413
	taud		2.109	2.012	1.863	1.862	1.909	1.984	2.077	2.102	2.104	2.096	2.158	2.162
	Ebn at Noon		245	245	239	240	239	238	242	244	241	237	238	241
	Edh at Noon		41	50	62	65	62	57	52	50	48	45	39	37
All-Sky Solar Radiation	RadAvg		859	1024	1315	1500	1533	1234	1689	1686	1363	1120	878	798
	RadStd		64	91	92	156	170	206	191	112	153	132	88	68

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (20 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page